

Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.5_e-1)

Metric

$$ds^2 = n dx \otimes dx + m dx \otimes dy + m dy \otimes dx + l dy \otimes dy + e^M dz \otimes dz - e^M dt \otimes dt$$

Coordinates

$$(x^\mu) = (x, y, z, t)$$

Metric Functions

$$n(t, z) = n$$

$$m(t, z) = m$$

$$l(t, z) = l$$

$$M(t, z) = M$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} i \sqrt{2} \sqrt{n} \right) dx + \left(\frac{1}{2} \sqrt{2} \sqrt{-\frac{m^2}{n}} + l + \frac{i \sqrt{2} m}{2 \sqrt{n}} \right) dy$$

$$n = \left(\frac{1}{2} i \sqrt{2} \sqrt{n} \right) dx + \left(-\frac{1}{2} \sqrt{2} \sqrt{-\frac{m^2}{n}} + l + \frac{i \sqrt{2} m}{2 \sqrt{n}} \right) dy$$

$$m = \left(\frac{1}{2} \sqrt{2} e^{(\frac{1}{2} M)} \right) dz + \left(\frac{1}{2} i \sqrt{2} \sqrt{-e^M} \right) dt$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} e^{(\frac{1}{2} M)} \right) dz + \left(-\frac{1}{2} i \sqrt{2} \sqrt{-e^M} \right) dt$$

Contravariant components

$$l = \left(-\frac{m^2}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} + \frac{ln}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} + \frac{\sqrt{m^2 - lnm}}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} \right) dx + \left(-\frac{\sqrt{m^2 - lnn}}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} \right) dy$$

$$n = \left(-\frac{m^2}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} + \frac{ln}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} - \frac{\sqrt{m^2 - lnm}}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} \right) dx + \left(\frac{\sqrt{m^2 - lnn}}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} \right) dy$$

$$m = \left(\frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} M)} \right) dz + \left(-\frac{1}{2} i \sqrt{2} \sqrt{-e^M} e^{(-M)} \right) dt$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} M)} \right) dz + \left(\frac{1}{2} i \sqrt{2} \sqrt{-e^M} e^{(-M)} \right) dt$$

Directional Derivatives

$$D = -\frac{m^2}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} + \frac{ln}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} + \frac{\sqrt{m^2 - lnm}}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} \partial_x - \frac{\sqrt{m^2 - lnn}}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} \partial_y$$

$$\Delta = -\frac{m^2}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} + \frac{ln}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} - \frac{\sqrt{m^2 - lnm}}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} \partial_x + \frac{\sqrt{m^2 - lnn}}{(\sqrt{2}m^2 - \sqrt{2}ln)\sqrt{-n}} \partial_y$$

$$\delta = \frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} M)} \partial_z - \frac{1}{2} i \sqrt{2} \sqrt{-e^M} e^{(-M)} \partial_t$$

$$\bar{\delta} = \frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} M)} \partial_z + \frac{1}{2} i \sqrt{2} \sqrt{-e^M} e^{(-M)} \partial_t$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\begin{aligned} \kappa = & -\frac{\sqrt{2} e^{(\frac{1}{2} M)} n^2 \frac{\partial}{\partial z} l}{8(e^M m^2 n - e^M l n^2)} + \frac{\sqrt{2} e^{(\frac{1}{2} M)} m n \frac{\partial}{\partial z} m}{4(e^M m^2 n - e^M l n^2)} - \frac{\sqrt{2} e^{(\frac{1}{2} M)} m^2 \frac{\partial}{\partial z} n}{4(e^M m^2 n - e^M l n^2)} + \frac{\sqrt{2} e^{(\frac{1}{2} M)} l n \frac{\partial}{\partial z} n}{8(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{-e^M} n^2 \frac{\partial}{\partial t} l}{8(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{-e^M} m n \frac{\partial}{\partial t} m}{4(e^M m^2 n - e^M l n^2)} \\ & - \frac{\sqrt{2} \sqrt{m^2 - l n} e^{(\frac{1}{2} M)} n \frac{\partial}{\partial z} m}{4(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{-e^M} m^2 \frac{\partial}{\partial t} n}{4(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{-e^M} l n \frac{\partial}{\partial t} n}{8(e^M m^2 n - e^M l n^2)} + \frac{\sqrt{2} \sqrt{m^2 - l n} e^{(\frac{1}{2} M)} m \frac{\partial}{\partial z} n}{4(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{m^2 - l n} \sqrt{-e^M} n \frac{\partial}{\partial t} m}{4(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{m^2 - l n} \sqrt{-e^M} m \frac{\partial}{\partial t} n}{4(e^M m^2 n - e^M l n^2)} \end{aligned}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & \frac{\sqrt{2} e^{(\frac{1}{2} M)} n^2 \frac{\partial}{\partial z} l}{8(e^M m^2 n - e^M l n^2)} - \frac{\sqrt{2} e^{(\frac{1}{2} M)} m n \frac{\partial}{\partial z} m}{4(e^M m^2 n - e^M l n^2)} + \frac{\sqrt{2} e^{(\frac{1}{2} M)} m^2 \frac{\partial}{\partial z} n}{4(e^M m^2 n - e^M l n^2)} - \frac{\sqrt{2} e^{(\frac{1}{2} M)} l n \frac{\partial}{\partial z} n}{8(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{-e^M} n^2 \frac{\partial}{\partial t} l}{8(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{-e^M} m n \frac{\partial}{\partial t} m}{4(e^M m^2 n - e^M l n^2)} \\ & - \frac{\sqrt{2} \sqrt{m^2 - l n} e^{(\frac{1}{2} M)} n \frac{\partial}{\partial z} m}{4(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{-e^M} m^2 \frac{\partial}{\partial t} n}{4(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{-e^M} l n \frac{\partial}{\partial t} n}{8(e^M m^2 n - e^M l n^2)} + \frac{\sqrt{2} \sqrt{m^2 - l n} e^{(\frac{1}{2} M)} m \frac{\partial}{\partial z} n}{4(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{m^2 - l n} \sqrt{-e^M} n \frac{\partial}{\partial t} m}{4(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{m^2 - l n} \sqrt{-e^M} m \frac{\partial}{\partial t} n}{4(e^M m^2 n - e^M l n^2)} \end{aligned}$$

$$\begin{aligned} \alpha = & -\frac{\sqrt{2} e^{(\frac{1}{2} M)} m^2 n \frac{\partial}{\partial z} M}{8(e^M m^2 n - e^M l n^2)} + \frac{\sqrt{2} e^{(\frac{1}{2} M)} l n^2 \frac{\partial}{\partial z} M}{8(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{-e^M} m^2 n \frac{\partial}{\partial t} M}{8(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{-e^M} l n^2 \frac{\partial}{\partial t} M}{8(e^M m^2 n - e^M l n^2)} - \frac{\sqrt{2} \sqrt{m^2 - l n} e^{(\frac{1}{2} M)} n \frac{\partial}{\partial z} m}{8(e^M m^2 n - e^M l n^2)} \\ & + \frac{\sqrt{2} \sqrt{m^2 - l n} e^{(\frac{1}{2} M)} m \frac{\partial}{\partial z} n}{8(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{m^2 - l n} \sqrt{-e^M} n \frac{\partial}{\partial t} m}{8(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{m^2 - l n} \sqrt{-e^M} m \frac{\partial}{\partial t} n}{8(e^M m^2 n - e^M l n^2)} \end{aligned}$$

$$\begin{aligned} \beta = & \frac{\sqrt{2} e^{(\frac{1}{2} M)} m^2 n \frac{\partial}{\partial z} M}{8(e^M m^2 n - e^M l n^2)} - \frac{\sqrt{2} e^{(\frac{1}{2} M)} l n^2 \frac{\partial}{\partial z} M}{8(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{-e^M} m^2 n \frac{\partial}{\partial t} M}{8(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{-e^M} l n^2 \frac{\partial}{\partial t} M}{8(e^M m^2 n - e^M l n^2)} - \frac{\sqrt{2} \sqrt{m^2 - l n} e^{(\frac{1}{2} M)} n \frac{\partial}{\partial z} m}{8(e^M m^2 n - e^M l n^2)} \\ & + \frac{\sqrt{2} \sqrt{m^2 - l n} e^{(\frac{1}{2} M)} m \frac{\partial}{\partial z} n}{8(e^M m^2 n - e^M l n^2)} + \frac{i \sqrt{2} \sqrt{m^2 - l n} \sqrt{-e^M} n \frac{\partial}{\partial t} m}{8(e^M m^2 n - e^M l n^2)} - \frac{i \sqrt{2} \sqrt{m^2 - l n} \sqrt{-e^M} m \frac{\partial}{\partial t} n}{8(e^M m^2 n - e^M l n^2)} \end{aligned}$$

$$\tau = \frac{\sqrt{2}e^{\left(\frac{1}{2}M\right)}n\frac{\partial}{\partial z}l}{8(e^Mm^2 - e^Mln)} - \frac{\sqrt{2}e^{\left(\frac{1}{2}M\right)}m\frac{\partial}{\partial z}m}{4(e^Mm^2 - e^Mln)} + \frac{\sqrt{2}e^{\left(\frac{1}{2}M\right)}l\frac{\partial}{\partial z}n}{8(e^Mm^2 - e^Mln)} - \frac{i\sqrt{2}\sqrt{-e^M}n\frac{\partial}{\partial t}l}{8(e^Mm^2 - e^Mln)} + \frac{i\sqrt{2}\sqrt{-e^M}m\frac{\partial}{\partial t}m}{4(e^Mm^2 - e^Mln)} - \frac{i\sqrt{2}\sqrt{-e^M}l\frac{\partial}{\partial t}n}{8(e^Mm^2 - e^Mln)}$$

$$\pi = -\frac{\sqrt{2}e^{\left(\frac{1}{2}M\right)}n\frac{\partial}{\partial z}l}{8(e^Mm^2 - e^Mln)} + \frac{\sqrt{2}e^{\left(\frac{1}{2}M\right)}m\frac{\partial}{\partial z}m}{4(e^Mm^2 - e^Mln)} - \frac{\sqrt{2}e^{\left(\frac{1}{2}M\right)}l\frac{\partial}{\partial z}n}{8(e^Mm^2 - e^Mln)} - \frac{i\sqrt{2}\sqrt{-e^M}n\frac{\partial}{\partial t}l}{8(e^Mm^2 - e^Mln)} + \frac{i\sqrt{2}\sqrt{-e^M}m\frac{\partial}{\partial t}m}{4(e^Mm^2 - e^Mln)} - \frac{i\sqrt{2}\sqrt{-e^M}l\frac{\partial}{\partial t}n}{8(e^Mm^2 - e^Mln)}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = & -\frac{n^3\frac{\partial}{\partial t}l^2}{16(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{m^2n^2\frac{\partial^2}{(\partial t)^2}l}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{ln^3\frac{\partial^2}{(\partial t)^2}l}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{n^3\frac{\partial}{\partial z}l^2}{16(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} \\ & + \frac{m^2n^2\frac{\partial^2}{(\partial z)^2}l}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{ln^3\frac{\partial^2}{(\partial z)^2}l}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{mn^2\frac{\partial}{\partial t}l\frac{\partial}{\partial t}m}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{m^2n\frac{\partial}{\partial t}m^2}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} \\ & + \frac{m^3n\frac{\partial^2}{(\partial t)^2}m}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{lmn^2\frac{\partial^2}{(\partial t)^2}m}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{mn^2\frac{\partial}{\partial z}l\frac{\partial}{\partial z}m}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{m^2n\frac{\partial}{\partial z}m^2}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} \\ & - \frac{m^3n\frac{\partial^2}{(\partial z)^2}m}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{lmn^2\frac{\partial^2}{(\partial z)^2}m}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{m^2n\frac{\partial}{\partial t}l\frac{\partial}{\partial t}n}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{m^3\frac{\partial}{\partial t}m\frac{\partial}{\partial t}n}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} \\ & - \frac{lm^2\frac{\partial}{\partial t}n^2}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{l^2n\frac{\partial}{\partial t}n^2}{16(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{m^4\frac{\partial^2}{(\partial t)^2}n}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} \\ & + \frac{3lm^2n\frac{\partial^2}{(\partial t)^2}n}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{l^2n^2\frac{\partial^2}{(\partial t)^2}n}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{m^2n\frac{\partial}{\partial z}l\frac{\partial}{\partial z}n}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{m^3\frac{\partial}{\partial z}m\frac{\partial}{\partial z}n}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} \\ & + \frac{lm^2\frac{\partial}{\partial z}n^2}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{l^2n\frac{\partial}{\partial z}n^2}{16(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{m^4\frac{\partial^2}{(\partial z)^2}n}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{3lm^2n\frac{\partial^2}{(\partial z)^2}n}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} \\ & + \frac{l^2n^2\frac{\partial^2}{(\partial z)^2}n}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{\sqrt{m^2 - l\bar{n}}n^2\frac{\partial}{\partial t}l\frac{\partial}{\partial t}m}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{\sqrt{m^2 - l\bar{n}}mn\frac{\partial}{\partial t}m^2}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{\sqrt{m^2 - l\bar{n}}m^2n\frac{\partial^2}{(\partial t)^2}m}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} \\ & + \frac{\sqrt{m^2 - l\bar{n}}ln^2\frac{\partial^2}{(\partial t)^2}m}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \frac{\sqrt{m^2 - l\bar{n}}n^2\frac{\partial}{\partial z}l\frac{\partial}{\partial z}m}{8(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} - \frac{\sqrt{m^2 - l\bar{n}}mn\frac{\partial}{\partial z}m^2}{4(e^Mm^4n - 2e^Mlm^2n^2 + e^Ml^2n^3)} + \dots\end{aligned}$$

$$\Phi_{01} = 0$$

$$\begin{aligned} \Phi_{02} = & -\frac{e^{(\frac{1}{2}M)}m^2n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}ln^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}m^2n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\ & + \frac{e^{(\frac{1}{2}M)}ln^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}m^3\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\ & - \frac{e^{(\frac{1}{2}M)}lmn\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}m^3\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}lmn\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\ & - \frac{e^{(\frac{1}{2}M)}lm^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}l^2n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}lm^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\ & + \frac{e^{(\frac{1}{2}M)}l^2n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{i\sqrt{-e^M}m^2n\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{i\sqrt{-e^M}ln^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\ & + \frac{i\sqrt{-e^M}m^2n\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{i\sqrt{-e^M}ln^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{i\sqrt{-e^M}m^3\frac{\partial}{\partial z}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\ & + \frac{i\sqrt{-e^M}lmn\frac{\partial}{\partial z}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{i\sqrt{-e^M}m^3\frac{\partial}{\partial t}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \dots \end{aligned}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{m^2\frac{\partial^2}{(\partial t)^2}M}{8(e^Mm^2-e^Mln)} - \frac{ln\frac{\partial^2}{(\partial z)^2}M}{8(e^Mm^2-e^Mln)} - \frac{m^2\frac{\partial^2}{(\partial z)^2}M}{8(e^Mm^2-e^Mln)} + \frac{ln\frac{\partial^2}{(\partial z)^2}M}{8(e^Mm^2-e^Mln)} - \frac{\frac{\partial}{\partial t}m^2}{16(e^Mm^2-e^Mln)} + \frac{\frac{\partial}{\partial z}m^2}{16(e^Mm^2-e^Mln)} + \frac{\frac{\partial}{\partial t}l\frac{\partial}{\partial t}n}{16(e^Mm^2-e^Mln)} - \frac{\frac{\partial}{\partial z}l\frac{\partial}{\partial z}n}{16(e^Mm^2-e^Mln)}$$

$$\Phi_{12} = 0$$

$$\begin{aligned}
\Phi_{20} = & -\frac{e^{(\frac{1}{2}M)}m^2n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}ln^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}m^2n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& + \frac{e^{(\frac{1}{2}M)}ln^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}m^3\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}lmn\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& + \frac{e^{(\frac{1}{2}M)}m^3\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}lmn\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}lm^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& + \frac{e^{(\frac{1}{2}M)}l^2n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}lm^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}l^2n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& - \frac{i\sqrt{-e^M}m^2n\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{i\sqrt{-e^M}ln^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{i\sqrt{-e^M}m^2n\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& + \frac{i\sqrt{-e^M}ln^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{i\sqrt{-e^M}m^3\frac{\partial}{\partial z}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& - \frac{i\sqrt{-e^M}lmn\frac{\partial}{\partial z}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{i\sqrt{-e^M}m^3\frac{\partial}{\partial t}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \dots
\end{aligned}$$

$$\Phi_{21} = 0$$

$$\begin{aligned}
\Phi_{22} = & -\frac{n^3 \frac{\partial}{\partial t} l^2}{16(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{m^2 n^2 \frac{\partial^2}{(\partial t)^2} l}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{ln^3 \frac{\partial^2}{(\partial t)^2} l}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{n^3 \frac{\partial}{\partial z} l^2}{16(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} \\
& + \frac{m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{ln^3 \frac{\partial^2}{(\partial z)^2} l}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{mn^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{m^2 n \frac{\partial}{\partial t} m^2}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} \\
& + \frac{m^3 n \frac{\partial^2}{(\partial t)^2} m}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{lmn^2 \frac{\partial^2}{(\partial t)^2} m}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{mn^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{m^2 n \frac{\partial}{\partial z} m^2}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} \\
& - \frac{m^3 n \frac{\partial^2}{(\partial z)^2} m}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{lmn^2 \frac{\partial^2}{(\partial z)^2} m}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{m^2 n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{m^3 \frac{\partial}{\partial t} m \frac{\partial}{\partial t} n}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} \\
& - \frac{lm^2 \frac{\partial}{\partial t} n^2}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{l^2 n \frac{\partial}{\partial t} n^2}{16(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{m^4 \frac{\partial^2}{(\partial t)^2} n}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} \\
& + \frac{3lm^2 n \frac{\partial^2}{(\partial t)^2} n}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{l^2 n^2 \frac{\partial^2}{(\partial t)^2} n}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{m^2 n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{m^3 \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} \\
& + \frac{lm^2 \frac{\partial}{\partial z} n^2}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{l^2 n \frac{\partial}{\partial z} n^2}{16(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{m^4 \frac{\partial^2}{(\partial z)^2} n}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{3lm^2 n \frac{\partial^2}{(\partial z)^2} n}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} \\
& + \frac{l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{\sqrt{m^2 - ln} n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{\sqrt{m^2 - ln} mn \frac{\partial}{\partial t} m^2}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{\sqrt{m^2 - ln} m^2 n \frac{\partial^2}{(\partial t)^2} m}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} \\
& - \frac{\sqrt{m^2 - ln} ln^2 \frac{\partial^2}{(\partial t)^2} m}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \frac{\sqrt{m^2 - ln} n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} + \frac{\sqrt{m^2 - ln} mn \frac{\partial}{\partial z} m^2}{4(e^M m^4 n - 2e^M l m^2 n^2 + e^M l^2 n^3)} - \dots
\end{aligned}$$

$$\begin{aligned}
\Lambda = & \frac{m^4 \frac{\partial^2}{(\partial t)^2} M}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{l m^2 n \frac{\partial^2}{(\partial t)^2} M}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} + \frac{l^2 n^2 \frac{\partial^2}{(\partial t)^2} M}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{m^4 \frac{\partial^2}{(\partial z)^2} M}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} \\
& + \frac{l m^2 n \frac{\partial^2}{(\partial z)^2} M}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{l^2 n^2 \frac{\partial^2}{(\partial z)^2} M}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{n^2 \frac{\partial}{\partial t} l^2}{48 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{m^2 n \frac{\partial^2}{(\partial t)^2} l}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} \\
& + \frac{l n^2 \frac{\partial^2}{(\partial t)^2} l}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} + \frac{n^2 \frac{\partial}{\partial z} l^2}{48 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} + \frac{m^2 n \frac{\partial^2}{(\partial z)^2} l}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} \\
& - \frac{l n^2 \frac{\partial^2}{(\partial z)^2} l}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} + \frac{m n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{m^2 \frac{\partial}{\partial t} m^2}{48 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{l n \frac{\partial}{\partial t} m^2}{16 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} \\
& + \frac{m^3 \frac{\partial^2}{(\partial t)^2} m}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{l m n \frac{\partial^2}{(\partial t)^2} m}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{m n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} + \frac{m^2 \frac{\partial}{\partial z} m^2}{48 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} \\
& + \frac{l n \frac{\partial}{\partial z} m^2}{16 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{m^3 \frac{\partial^2}{(\partial z)^2} m}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} + \frac{l m n \frac{\partial^2}{(\partial z)^2} m}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{m^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{16 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} \\
& + \frac{l n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{48 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} + \frac{l m \frac{\partial}{\partial t} m \frac{\partial}{\partial t} n}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{l^2 \frac{\partial}{\partial t} n^2}{48 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{l m^2 \frac{\partial^2}{(\partial t)^2} n}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} \\
& + \frac{l^2 n \frac{\partial^2}{(\partial t)^2} n}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} + \frac{m^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{l n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{48 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{l m \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{12 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} \\
& + \frac{l^2 \frac{\partial}{\partial z} n^2}{48 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} + \frac{l m^2 \frac{\partial^2}{(\partial z)^2} n}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)} - \frac{l^2 n \frac{\partial^2}{(\partial z)^2} n}{24 (e^M m^4 - 2 e^M l m^2 n + e^M l^2 n^2)}
\end{aligned}$$

Weyl Tensor Components

$$\begin{aligned}
\Psi_0 = & -\frac{e^{(\frac{1}{2}M)}m^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} + \frac{e^{(\frac{1}{2}M)}ln^3\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}m^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& + \frac{e^{(\frac{1}{2}M)}ln^3\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} + \frac{e^{(\frac{1}{2}M)}m^3n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}lmn^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& + \frac{e^{(\frac{1}{2}M)}m^3n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}lmn^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}m^4\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& + \frac{3e^{(\frac{1}{2}M)}lm^2n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}l^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}m^4\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& + \frac{3e^{(\frac{1}{2}M)}lm^2n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}l^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} + \frac{i\sqrt{-e^M}m^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& - \frac{i\sqrt{-e^M}ln^3\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} + \frac{i\sqrt{-e^M}m^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{i\sqrt{-e^M}ln^3\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \dots
\end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned}
\Psi_2 = & -\frac{e^{(\frac{1}{2}M)}m^4\frac{\partial^2}{(\partial t)^2}M}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}lm^2n\frac{\partial^2}{(\partial t)^2}M}{6\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}l^2n^2\frac{\partial^2}{(\partial t)^2}M}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& + \frac{e^{(\frac{1}{2}M)}m^4\frac{\partial^2}{(\partial z)^2}M}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}lm^2n\frac{\partial^2}{(\partial z)^2}M}{6\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}l^2n^2\frac{\partial^2}{(\partial z)^2}M}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& - \frac{e^{(\frac{1}{2}M)}n^2\frac{\partial}{\partial t}l^2}{48\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}m^2n\frac{\partial^2}{(\partial t)^2}l}{24\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}ln^2\frac{\partial^2}{(\partial t)^2}l}{24\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& + \frac{e^{(\frac{1}{2}M)}n^2\frac{\partial}{\partial z}l^2}{48\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}m^2n\frac{\partial^2}{(\partial z)^2}l}{24\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& - \frac{e^{(\frac{1}{2}M)}ln^2\frac{\partial^2}{(\partial z)^2}l}{24\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \frac{e^{(\frac{1}{2}M)}mn\frac{\partial}{\partial t}l\frac{\partial}{\partial t}m}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}m^2\frac{\partial}{\partial t}m^2}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& + \frac{e^{(\frac{1}{2}M)}m^3\frac{\partial^2}{(\partial t)^2}m}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}lmn\frac{\partial^2}{(\partial t)^2}m}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}mn\frac{\partial}{\partial z}l\frac{\partial}{\partial z}m}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} \\
& + \frac{e^{(\frac{1}{2}M)}m^2\frac{\partial}{\partial z}m^2}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} - \frac{e^{(\frac{1}{2}M)}m^3\frac{\partial^2}{(\partial z)^2}m}{12\left(e^{(\frac{3}{2}M)}m^4-2e^{(\frac{3}{2}M)}lm^2n+e^{(\frac{3}{2}M)}l^2n^2\right)} + \dots
\end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}
\Psi_4 = & -\frac{e^{(\frac{1}{2}M)}m^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} + \frac{e^{(\frac{1}{2}M)}ln^3\frac{\partial}{\partial t}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}m^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& + \frac{e^{(\frac{1}{2}M)}ln^3\frac{\partial}{\partial z}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} + \frac{e^{(\frac{1}{2}M)}m^3n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}lmn^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}m}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& + \frac{e^{(\frac{1}{2}M)}m^3n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}lmn^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}m}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}m^4\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& + \frac{3e^{(\frac{1}{2}M)}lm^2n\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}l^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial t}n}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}m^4\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{4\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& + \frac{3e^{(\frac{1}{2}M)}lm^2n\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{e^{(\frac{1}{2}M)}l^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial z}n}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{i\sqrt{-e^M}m^2n^2\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} \\
& + \frac{i\sqrt{-e^M}ln^3\frac{\partial}{\partial z}M\frac{\partial}{\partial t}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} - \frac{i\sqrt{-e^M}m^2n^2\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} + \frac{i\sqrt{-e^M}ln^3\frac{\partial}{\partial t}M\frac{\partial}{\partial z}l}{8\left(e^{(\frac{3}{2}M)}m^4n-2e^{(\frac{3}{2}M)}lm^2n^2+e^{(\frac{3}{2}M)}l^2n^3\right)} + \dots
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"